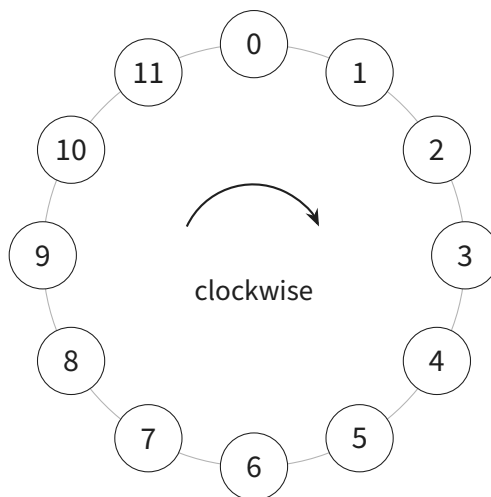


Problem 1: A shift $+k$ moves k places clockwise, wrapping after 11. Try $+4$ from 0, then from 9. Record both landings. Find where inputs 2 and 8 go under the same shift.



Problem 2: A secret shift takes 9 to 1. Use the ring to recover it. Test the additional clue 2 to 6. Decode outputs 1,4,10 by undoing that shift.

Problem 3: Repeat $+4$ from 0 until it first returns. Write the whole route. Start at a position not yet used and repeat. Continue until every position is in a recorded loop.

Problem 4: Use the twelve-place ring. Start at 0 and repeat +8 until you return. Find its other loops. Compare the loop members and their order with +4.

Problem 5: Try +3, then +6, then +5. For each rule, find the route from 0 and all remaining loops. Record the first return time and the number of loops.

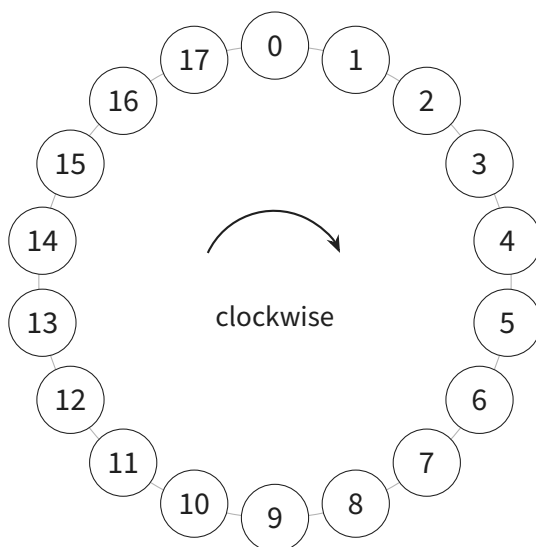
Shift	Routes	Return	Count
+3			
+6			
+5			

Problem 6: Compare these shifts with +4 and +8. Group shifts with the same return time. Record a prediction connecting the step size, ring size, and number of loops.

Problem 7: Test the remaining shifts on the twelve-place ring. Record first positive return times and number of loops. Use your earlier results where you have them. For +0, a turn leaves the counter in place.

Shift	Return	Loops	Shift	Return	Loops
0			6		
1			7		
2			8		
3			9		
4			10		
5			11		

Problem 8: On this eighteen-place ring, predict the first return for +6 and +5. Test both and record their routes from 0. Compare with your prediction.



Problem 9: Each +4 turn on twelve places moves four spaces. Fill the distance table and circle the first positive total that is a whole number of laps. Repeat for +8. Compare this distance record with the routes you drew.

Turns	+4 distance	Whole laps?	+8 distance	Whole laps?
1				
2				
3				
4				

Problem 10: Find a shift that undoes +4 on twelve places. Test it from 0,1,5, and 11. Record the four out-and-back trips.

Problem 11: Start at 0, then 2, then 9. Compare +4 followed by +7 with +7 followed by +4. Record both final landings for each start. Find one shift that does the same job.

Problem 12: Choose another pair of shifts on twelve places. Test both orders from two starts. Record your combined shift and a rule suggested by the experiments.